

$$(A^{-1})^* = (A^*)^{-1}$$

Tipos de matrices:

- Matriz hermitica: $H^* = H$

$$\therefore H \in \mathbb{R}^{n \times n} \Rightarrow H = H^* \quad \bar{h}_{ii} = h_{ii} \in \mathbb{R} \quad i=1, \dots, n$$

- Matriz Unitarias $\Rightarrow U^* = U^{-1}$

s: $U \in \mathbb{R}^{n \times n}$ ortogonal (autoconjugada)

$$U \cdot U^* = I = U^* \cdot U = U^* U = \begin{pmatrix} u_1^* \\ \vdots \\ u_n^* \end{pmatrix} (u_1 \dots u_n) = \begin{pmatrix} \underbrace{u_1^* u_1}_{\|u_1\|^2} & \dots & \underbrace{u_1^* u_n}_{\|u_1\|^2} \\ \vdots & & \vdots \\ \underbrace{u_n^* u_1}_0 & \dots & \underbrace{u_n^* u_n}_{\|u_n\|^2} \end{pmatrix} \Rightarrow \begin{aligned} \|u_i\|_2^2 &= 1 \quad \forall i=1, \dots, n \\ \langle u_i, u_j \rangle &= 0 \quad i \neq j \end{aligned}$$

$$u^* u = (\bar{u}_1 \dots \bar{u}_n) \begin{pmatrix} u_1 \\ \vdots \\ u_n \end{pmatrix} = \sum_{i=1}^n \bar{u}_i u_i = \|u\|^2$$

- Matrices normales $AA^* = A^*A$

hermitica $\Rightarrow$ normal

unitarias $\Rightarrow$ normales

$$UU^* = I = U^*U$$

- Matrices diagonales $D = \text{diag}(d_1, \dots, d_n) \quad d_{ij} = 0 \quad \forall i \neq j$

- Matrices triangulares inferior/superior

Ej: A, B matrices triang. inf. $\Rightarrow C = AB$ triang. inf.

Dem:

$$A_n = \left(\begin{array}{c|c} c_{11} & 0^t \\ \hline \alpha & A_{n-1} \end{array} \right) \quad \left(\begin{array}{c|c} b_{11} & 0^t \\ \hline \beta & B_{n-1} \end{array} \right) = \left(\begin{array}{c|c} \underbrace{c_{11}b_{11}}_{C_{11}} & 0^t \\ \hline \alpha b_{11} + A_{n-1}\beta & A_{n-1}B_{n-1} \end{array} \right)$$

$\uparrow$
 triang. inf.

$\underbrace{\hspace{10em}}_C \quad \underbrace{\hspace{10em}}_{C_{n-1}}$

$$C_{ii} = a_{ii}b_{ii}$$

- Matriz banda.

$$A = \begin{pmatrix} & & 0 \\ & \diagdown & \\ 0 & & \end{pmatrix}$$

- Matriz de diagonal estricta dominante (d.e.d)

$$A \in \mathbb{R}^{n \times n} \quad |a_{ii}| > \sum_{j=1, j \neq i}^n |a_{ij}|$$

Teorema:

$$A \in \mathbb{R}^{n \times n}$$

$$\text{tr} : \mathbb{R}^{n \times n} \rightarrow \mathbb{R}$$

$$A \mapsto \text{tr}(A) = \sum_{i=1}^n a_{ii}$$

$$\begin{cases} x_0 = 1 \\ x_1 = 1/3 \\ x_{n+1} = \frac{2}{3}x_n - \frac{1}{3}x_{n-1} \end{cases}$$

$$x_n = p^n C$$

$$p^{n+1} C = \frac{2}{3} p^n C - \frac{1}{3} p^{n-1} C$$

$$\Rightarrow p^2 - \frac{2}{3}p + \frac{1}{3} = 0 \quad \leadsto p = \{3, 1/3\}$$

La solución general $x_n = C_1 \cdot 3^n + C_2 \cdot 1/3^n$

$$\Rightarrow \mu^2 - \frac{26}{7}\mu + 3/7 = 0 \quad \leadsto \mu = \{3, 1/3\}$$

La solución general $x_n = c_1 \cdot 3^n + c_2 \cdot 1/3^n$

$$x_0 = c_1 + c_2 = 1$$

$$x_1 = c_1 \cdot 3 + c_2 \cdot 1/3 = 1/3$$

$$27c_1 + c_2 = 1$$

Propiedades traza:

$$\bullet \operatorname{tr}(A+B) = \operatorname{tr}(A) + \operatorname{tr}(B)$$

$$\bullet \operatorname{tr}(A^T) = \operatorname{tr}(A)$$

$$\bullet \operatorname{tr}(\underbrace{AB}_C) = \operatorname{tr}(\underbrace{BA}_D) \quad A \in \mathbb{K}^{n \times p}, B \in \mathbb{K}^{p \times n}$$

$$c_{ii} = \sum_{k=1}^p a_{ik} \cdot b_{ki} \quad \operatorname{tr}(C) = \sum_{i=1}^n \sum_{k=1}^p a_{ik} \cdot b_{ki} = \sum_{k=1}^p \sum_{i=1}^n b_{ki} \cdot a_{ik}$$

$$d_{kk} = \sum_{i=1}^n b_{ki} \cdot a_{ik}$$

Determinante:

$$\det: \mathbb{K}^{n \times n} \rightarrow \mathbb{K}$$

$$A \mapsto \det(A) = \sum_{\sigma \in G_n} \overset{\text{signature}}{\operatorname{sig}(\sigma)} \cdot a_{1\sigma(1)} \cdot a_{2\sigma(2)} \cdots a_{n\sigma(n)}$$

$$G_n \equiv \text{grupo de permutaciones } \{1, 2, \dots, n\}$$

$$\sigma \in G_n \text{ permutación } \sigma = \{2, 1, 3\} \quad |G_n| = n!$$

$$\operatorname{sig}(\sigma) = (-1)^{N_\sigma}$$

$$N_\sigma \equiv \# \text{ cambios de orden } (i, j) \in \{1, \dots, n\} \quad j < i, \quad \sigma(i) < \sigma(j)$$

$$\sigma = \{2, 4, 5, 1, 3\} \leadsto 5 \text{ cambios} \quad N_\sigma = 5 \Rightarrow \operatorname{sig}(\sigma) = -1$$

Propiedades:

$$1) \det(\lambda A) = \lambda^n \det(A)$$

$$2) \det(A) = \prod_{i=1}^n \lambda_{ii} \quad D \text{ diagonal}$$

$$3) \det(T) = \prod_{i=1}^n t_{ii}$$

Prop.: $U \in \mathbb{K}^{n \times n}$ unitaria, entonces $|\det(U)| = 1$

Demos.:

$$U U^* = I = U U^*$$

$$\det(U U^*) = \det(U) \cdot \det(U^*) = \det(U) \cdot \det(\bar{U}) = |\det(U)|^2 = \det(I) = 1$$

Teorema: sea $A \in \mathbb{K}^{n \times n}$. Entonces son equivalentes las siguientes afirmaciones

$$1) \exists A^{-1}$$

$$2) \det(A) \neq 0$$

$$3) 0 \notin \operatorname{sp}(A)$$

Demostración:

① Supongamos que $\exists A^{-1}$ vemos que $\det(A) \neq 0$

$$a) \det(A) < \infty \quad (\text{sumatorio de productos finitos})$$

$$|\det(A^{-1})| < \infty$$

$$AA^{-1} = I \quad \det(A \cdot A^{-1}) = \det(I) = 1 \quad \Rightarrow \quad \det(A) = (\det(A^{-1}))^{-1} \neq 0.$$

②

$$\det(A) \neq 0 \Rightarrow \exists A^{-1} \quad (\text{más adelante}).$$

$\forall A \in \mathbb{K}^{n \times n} \quad \exists \mathbb{K}^{n \times n}$ tal que es invertible la matriz este compuesto del producto

$$E = E_1 \cdot E_2 \cdot \dots \cdot E_r$$

matrices elementales

$$EA = D = \text{diag}(d_1, \dots, d_n) \quad d_{ii} \in \{0, 1\}$$

$$\det(EA) = \det(D) = \prod_{i=1}^n d_{ii} = \frac{\det(E)}{\neq 0} \cdot \frac{\det(A)}{\neq 0} \Rightarrow \det(D) = \prod_{i=1}^n d_{ii} \neq 0 \Rightarrow d_{ii} = 1 \quad \forall i = 1, \dots, n \Rightarrow D = I$$

$$\Rightarrow E = A^{-1}$$

Autovalores y Autovectores de una matriz.

$$A \in \mathbb{K}^{n \times n} \quad \text{guamos } A^n.$$

Ejemplo: $ED \quad e^A := \sum_{n=0}^{\infty} \frac{A^n}{n!}$

$$Av = \lambda v \quad \sim \quad Av - \lambda v = 0 \Rightarrow (A - \lambda I)v = 0 \quad \Rightarrow \quad \det(A - \lambda I) = 0$$

$$0 = \begin{vmatrix} a_{11} - \lambda & \dots & a_{1n} \\ \vdots & \ddots & \vdots \\ a_{n1} & \dots & a_{nn} - \lambda \end{vmatrix} = P(\lambda) \in \mathbb{P}_n$$

polinomio característico

$$P(\lambda) = 0 \quad \xrightarrow{T \neq A} \quad \text{hay } n \text{ raíces complejas}$$

$$D = \begin{pmatrix} \lambda_1 & 0 & \dots & 0 \\ 0 & \lambda_2 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & \lambda_n \end{pmatrix} = \begin{pmatrix} \lambda_1 & & \\ & \ddots & \\ & & \lambda_n \end{pmatrix} \cdot I$$

Autovectores:

$$\lambda_i: (A - \lambda_i I) v_i = 0 \Rightarrow v_i$$

$$\det \left(\begin{array}{c|c} A & C \\ \hline I & B \end{array} \right) = \det \left(\left(\begin{array}{c|c} A & C \\ \hline I & I \end{array} \right) \begin{pmatrix} I \\ B \end{pmatrix} \right) = \det \left(\begin{array}{c|c} A & C \\ \hline I & I \end{array} \right) \cdot \det \left(\begin{pmatrix} I \\ B \end{pmatrix} \right) = \det(A) \cdot \det(B)$$

$$\det \left(\begin{array}{c|c} A & C \\ \hline I & B \end{array} \right)$$

$$\left(\begin{array}{c|c} I & T \\ \hline I & I \end{array} \right) \left(\begin{array}{c|c} A & C \\ \hline I & B \end{array} \right) = \left(\begin{array}{c|c} A & C + TB \\ \hline I & B \end{array} \right) \quad \xrightarrow{=0} \quad C + TB = 0 \Leftrightarrow T = -CB^{-1}$$

podemos escoger $T = -CB^{-1}$ porque la matriz T no altera $\det \left(\begin{pmatrix} I & T \\ I & I \end{pmatrix} \right)$

$$\Rightarrow \det(A) \cdot \det(B) = \det \left(\begin{array}{c|c} A & C \\ \hline I & B \end{array} \right) = \det \left(\begin{array}{c|c} I & T \\ \hline I & I \end{array} \right) \cdot \det \left(\begin{array}{c|c} A & C \\ \hline 0 & B \end{array} \right)$$

$$\text{¿qué ocurre si no existe } B^{-1}? \Rightarrow \det(B) = 0$$

Def: Sea $A \in \mathbb{K}^{n \times n}$ y $\text{sp}(A) = \{ \lambda_i \}_{i=1}^n$ entonces el radio espectral de A es:

$$\rho = \max_{1 \leq i \leq n} |\lambda_i|$$

P12

a) $\text{sp}(AB) = \text{sp}(BA)$

Demostración

1) $\lambda = 0 \in \text{sp}(AB) \Rightarrow 0 = \det(AB) = \det(A) \cdot \det(B) = \det(B) \cdot \det(A) \Rightarrow 0 \in \text{sp}(BA)$

2) $\lambda \neq 0 \in \text{sp}(AB) \Rightarrow \exists v \neq 0 \text{ q } ABv = \lambda v \neq 0 \Rightarrow BABv = \lambda Bv \Rightarrow BAu = \lambda u$
 $u = Bv \neq 0$. p.s. si no como $ABv = \lambda v \Rightarrow \lambda v = 0$ ✓

b) $A \in M_n(\mathbb{K}) \sim P(A)$

$$A^n = \underbrace{A \cdot A \cdot \dots \cdot A}_{n \text{ veces}} \quad \text{¿} P(A^n) \text{?}$$

$k \in \mathbb{N}$.

$$A^k u = \mu u \quad \mu \in \text{Sp}(A^k) \quad Av = \lambda v \quad \lambda \in \text{Sp}(A)$$

$$k=2 \quad Av = \lambda v \quad \leadsto \quad A^2 v = \lambda Av = \lambda(\lambda v) = \lambda^2 v$$

$$\leadsto A^k v = \lambda^k v \quad \leadsto \quad \text{Sp}(A^k) = \{ \lambda^k : \lambda \in \text{Sp}(A) \}$$

$$A = P^{-1} D P$$